



COMPUTER PROGRAMMING

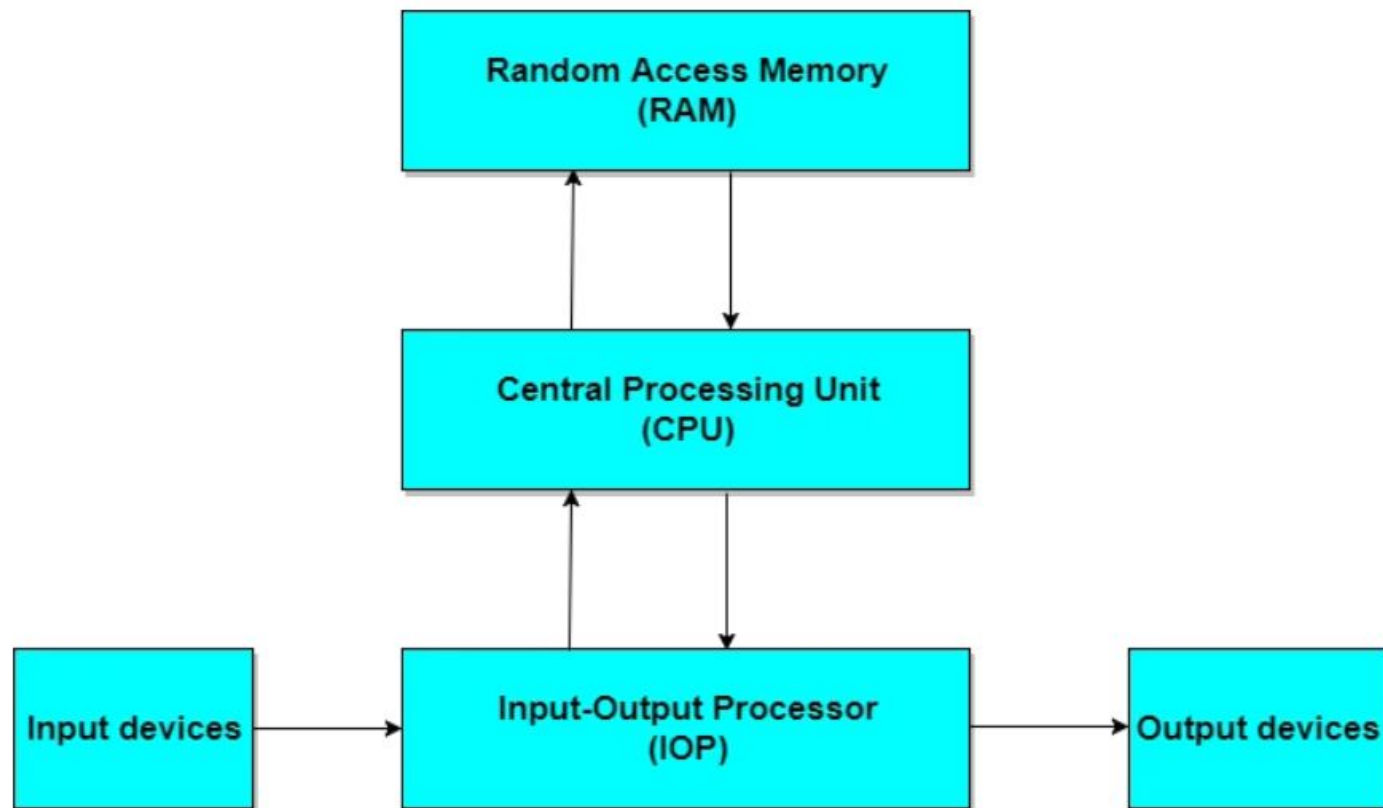
Introduction

- A computer is ‘an automatic electronic apparatus for making calculations or controlling operations that are expressible in numerical or logical terms’.
- The definition points towards the two major areas of computer application: data processing and computer-assisted controls or operations.
- Another important conclusion of the definition is the fact that the computer can perform only those operations or calculations that can be expressed in logical or numerical terms.

Digital Computers

- The von Neumann architecture describes a general framework, or structure, that a computer's hardware, programming, and data should follow.
- Although other structures for computing have been devised and implemented, the vast majority of computers in use today operate according to the von Neumann architecture.
- von Neumann envisioned the structure of a computer system as being composed of the following components:
- ALU: The Arithmetic-Logic unit that performs the computer's computational and logical functions.
- RAM: The computer's main, or fast, memory, also known as Random Access Memory(RAM).
- Control Unit: This is a component that directs other components of the computer to perform certain actions, such as directing the fetching of data or instructions from memory to be processed by the ALU.
- Man-machine interfaces; i.e. input and output devices, such as keyboard for input and display monitor for output.

Block diagram of a Digital Computer



Classification of Computers

- Most designs of computers today are based on concepts developed by John von Neumann and are referred to as the von Neumann architecture.
- Computers can be classified in variety of ways on the basis of various parameters such as usage, cost, size, processing power, and so on.
- The classification of computers is presented below based on their power and their use.

- Supercomputer

- Supercomputer is the most expensive and fastest type of computer.
- These are employed for specialized applications that require immense amounts of mathematical calculations such as weather forecasting, nuclear energy research, and petroleum exploration etc.

- Mainframe

- A mainframe computer supports a vast number of users to work simultaneously and remotely.
- It can process large amounts of data at very high speeds and support many input, output and auxiliary storage devices.
- These computers are very large in size, and expensive.
- Supercomputer can execute a single program faster than a mainframe, whereas a mainframe uses its power to execute many programs concurrently.

- Minicomputers

- A minicomputer is powerful enough to be used by multiple users (between 10 to 100) but is smaller in size and memory capacity and cheaper than mainframes.

- Microcomputers

- The microcomputer has been intended to meet the personal computing needs of an individual.

- Desktop computer

- Laptop computer

- Palmtop computer/Digital diary/Notebook/PDAs

- A hand sized microcomputer having no keyboard. The screen serves both as an input and output device.

ANATOMY OF A COMPUTER

- Hardware is the physical components of a computer that includes all mechanical, electrical, electronic and magnetic parts attached to it.
- A computer consists of the following major hardware components:
 - Input and output devices
 - Central processing unit (CPU)
 - Memory unit and storage devices
 - Interface unit

I/O Devices

- Keyboard.
- A keyboard, normally, consists of 104 keys. These keys are classified into different categories.
- Character keys: These keys include letters, numbers, and punctuation marks.
- Function keys : There are 12 functional keys above the key board which are used to perform certain functions depending on the operating system or the software currently being executed. These keys are placed at the top of the key board and can easily be identified with the letter F followed by a number ranging from 1 to 12.
- Control keys: Alt, Ctrl, Shift, Insert, Delete, Home, End, PgUp, PgDn, Esc and Arrow keys are control keys.
- Navigation keys: These include four arrows, Page Up and Page Down, Home and End. These keys are normally used to navigate around a document or screen.

- Toggle keys: Scroll Lock, Num lock, Caps Lock are three toggle keys.
- Miscellaneous keys: These keys include Insert, delete, escape, print Screen etc.
- The keys on the keyboard are placed in a series of rows and columns called the key matrix.
- Each key holds a position with respect to a row and column.
- When a key is pressed, the key switch in that position closes a circuit, sending a signal to the circuit board inside the keyboard.
- The keyboard controller uses the x and y coordinates of the matrix position to determine which key was pressed, thereby determining what code is transmitted to the computer by the keyboard.

- Mouse

- A mouse is the pointing device attached to a computer.
- The mouse tracks the motion of the mouse pointer and senses the clicks and sends them to the computer so it can respond appropriately.
- The mouse can be connected to the system either through a USB connector or wirelessly through infrared radiation. A wireless mouse needs to be powered through batteries.

- Scanner

- A scanner is a device that captures pictures or documents so that they can be stored in storage devices, seen on the video screen, modified suitably, transported to other computers, or printed on a printer

- Monitor
- Technologies- CRT, LCD, OLED
- Printer
- An impact printer is one where the print head will be in physical contact with the paper. Eg: Dot – matrix printer
- In a non–impact printer, on the other hand the print head will have no physical contact with the paper. Eg: Laser printer.

CPU

- Central Processing Unit or CPU can be thought of as the brain of the computer.
- Most of the processing takes place in CPU.
- During processing, it locates and executes the program instructions. It also fetches data from memory and input/output devices and sends data back.
- Physically, it is an integrated circuit (IC) silicon chip, mounted on a small square plastic slab, surrounded by metal pins.
- In the world of personal computers, the term microprocessor and CPU are used interchangeably.
- The CPU itself can be divided into different functional units :
- Registers: These are high-speed storage devices. In most CPUs, some registers are reserved for special purposes.
 - Instruction Register (IR) holds the current instruction being executed.
 - The Program Counter (PC) is a register that holds the address of the next instruction to be executed.
 - In addition to such and other special-purpose registers, it also contains a set of general-purpose registers that are used for temporary storage of data values as needed during processing.

- Arithmetic logic unit (ALU) It is the part of the CPU that performs arithmetic operations, such as addition and subtraction as well as logical operations, such as comparing two numbers to see if they are the equal or greater or less.
- Control unit (CU) The control unit coordinates the processing by controlling the transfer of data and instructions between main memory and the registers in the CPU.
- It also coordinates the execution of the arithmetic logic unit (ALU) to perform operations on data stored in particular registers.
- Nowadays, a high-speed memory, called cache memory, is embedded with the CPU chip. This improves the computer performance by minimizing the processor need to read data from the slow main memory.

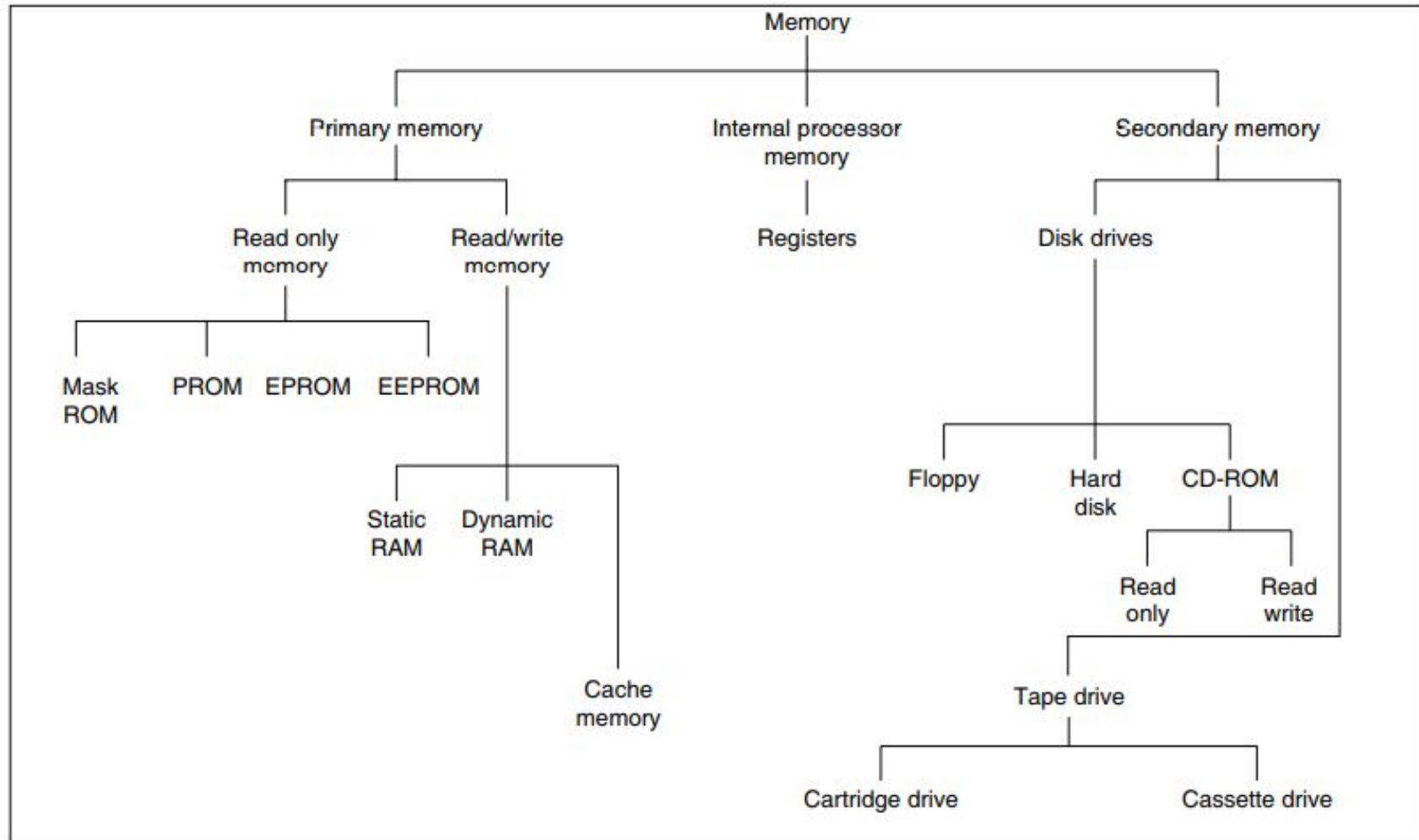
- The CPU's processing power is measured in terms of the number of instructions that it can execute per unit time.
- Every computer comprises of an internal clock, which emits electronic pulses at a constant rate.
- These pulses are used to control and synchronize the pace of operations. Each pulse is called a clock cycle.
- Nowadays, multiple processors are embedded together on a single integrated-circuit chip, known as multi-core processor e.g. a dual-core processor has two CPUs and a quad core processor has four CPUs

Memory unit

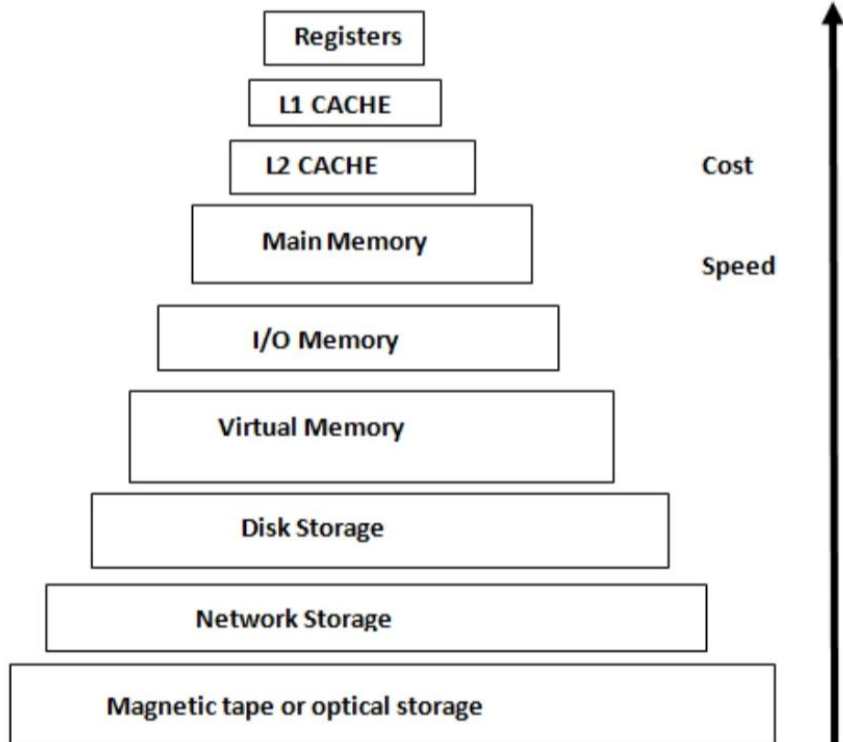
- Data are stored in memory as binary digits, called bits.
- Data of various types, such as numbers, characters, are encoded as series of bits and stored in consecutive memory locations.
- Each memory location comprises of a single byte which is equal to eight bits and has a unique address so that the contents of the desired memory locations can be accessed independently by referring to its' address.
- A single data item is stored in one or more consecutive bytes of memory. The address of the first byte is used as the address of the entire memory location.

- There are mainly two types of memory that are used in a computer system. One is called primary memory and the other secondary memory.
- Primary memory is the area where data and programs are stored while the program is being executed along with the data. from top to bottom, the speed decreases while the capacity increases and the prices become much lower
- This memory space, also known as main memory, forms the working area of the program. This memory is accessed directly by the processor.
- These type are called RAM, where any location can be accessed directly, and its stored contents get destroyed the moment power to this module is switched off.
- Hence, these are volatile in nature. Primary memory devices are expensive.
- They are limited in size, consume very low power, and are faster as compared to secondary memory devices

- There is another kind of primary memory increasingly being used in modern computers. It is called cache memory.
- It is a type of high-speed memory that allows the processor to access data more rapidly than from memory located elsewhere on the system.
- Secondary memory Secondary memory provides large, non-volatile, and inexpensive storage for programs and data.
- However, the access time in secondary memory is much larger than in primary memory.
- Secondary storage permits the storage of computer instructions and data for long periods of time.
- Moreover, secondary memory, which is also known as auxiliary memory, stores a huge number of data bytes at a lesser cost than primary memory devices.



- Memory operations
- There are some operations common to both primary and secondary memory devices. These are as follows.
 - *Read* During this operation, data is retrieved from memory.
 - *Write* In this operation, data is stored in the memory.
- The smallest block of memory is considered to be a byte, which comprises eight bits.
 - Kilobyte (KB) = 1024 bytes
 - Megabyte (MB) = 1024 Kilobytes
 - Gigabyte (GB) = 1024 Megabytes
 - Terabyte (TB) = 1024 Gigabytes
 - Petabyte (PB) = 1024 Terabytes
 - Exabyte (EB) = 1024 Petabytes
 - Zettabyte (ZB) = 1024 Exabytes
 - Yottabyte (YB) = 1024 Zettabytes

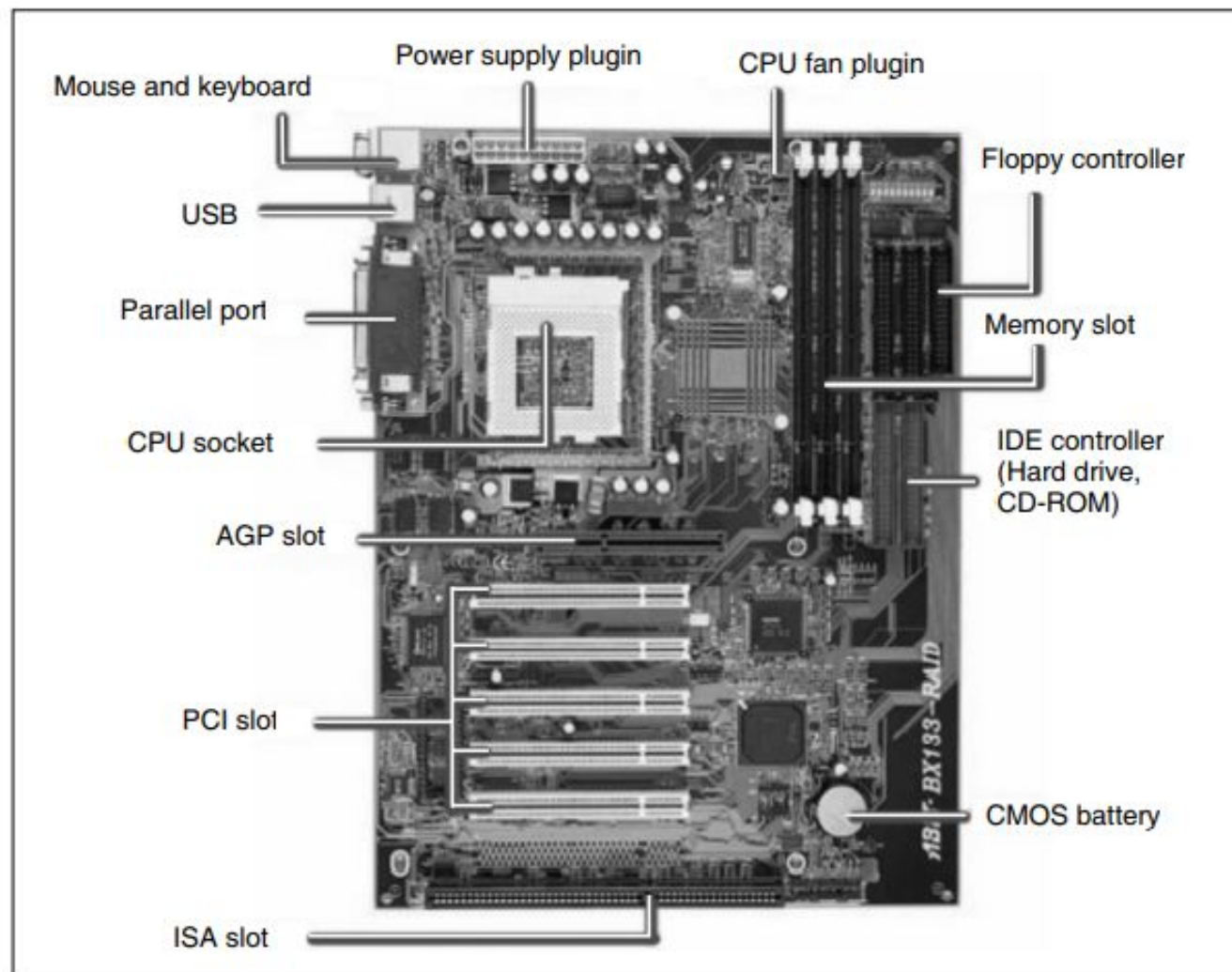


- From top to bottom, the speed decreases while the capacity increases and the prices become much lower.
- Those above the secondary memory are volatile and the rest are non-volatile.

Interface unit

- The interface unit interconnects the CPU with memory and also with the various (I/O) devices.
- The instructions and data move between the CPU and other hardware components through interface unit.
- It is a set of parallel wires or lines which connects all the internal computer components to the CPU and main memory.
- Depending on the type of data transmitted, a bus can be classified into the following three types:
- *Data bus* The bus used to carry actual data.
- *Address bus* memory or Input/output device Addresses travel via the address bus.
- *Control bus* This bus carries control information between the CPU and other devices within the computer. The control information entails signals that report the status of various devices, or ask devices to take specific actions.

- Motherboard
- All the components in the computer system are mounted and connected together by an electronic circuit board called motherboard or main board.
- To make all these things work together the motherboard provides some kind of physical connection among them



- *CPU socket* This holds the central processor which is an integrated chip along with the system clock, cache, cooling fan, etc.
- *Memory sockets* These sockets hold the RAM card that contains RAMs.
- *Interface module* This is for the hard disk, floppy disk, and CD-ROM drives.
- *ROM integrated chip* This is embedded with the basic input/output system software.

- *Ports and expansion slots* A port is used to connect a device with the bus. Physical ports include serial and parallel ports, to which peripheral devices such as printers, scanners, and external storage devices can be attached.
- The slots are used to attach accessories such as graphics (video) cards, disk controllers, and network cards.
- There are two different standards for expansion slots: ISA (Industry Standard Architecture) and PCI (Peripheral Component Interconnect).

- *ISA slots* These are for connecting ISA compatible cards.
- *PCI slots* These are for connecting I/O devices.
- *Advanced graphics port (AGP)* Video card is inserted into this slot.
- *Parallel port* The parallel port is also known as the printer port, or LPT1. It is capable of sending eight bits of information at a time.
- *Serial ports* these are sometimes called communication ports or COM ports. There are two COM ports, COM1 and COM2. size of COM1 is larger than that of COM2. COM1 has 25 pins and used for connecting Modems. COM2 is 9 pin port used for interfacing serial mouse.
- *USB (universal serial bus)* This is also a serial port but data rate is more than the serial port. USB is used as a general purpose communication channel in Personal Computers. Many different devices, such as mouse, keyboards, hard disk drives, portable CD-ROM/DVD drives, pen-drives, scanners, cameras, modems and even printers are usually connected to these ports.

- CMOS
- The CMOS in a Personal Computer stands for Complementary Metal Oxide Semiconductor memory.
- It is a type of RAM that stores the necessary attributes of system components, such as the size of the hard disk, the amount of RAM, and the resources used by the serial and parallel ports etc.

Software

- Software provides the instructions that tell the hardware exactly what is to be performed and in what order.
- This set of instructions is sequenced and organized in a computer program.
- Therefore, a program is a series of instructions which is intended to direct a computer to perform certain functions and is executed by the processor.
- In a broader sense, software can be described as a set of related programs. But software is more than a collection of programs.

- *System software* is designed to facilitate and coordinate the use of the computer by making hardware operational. It interacts with the computer system at low level. Examples of such software include language translator, operating system, loader, linker, etc.
- *Application software* is designed to perform specific usages of the users. Microsoft Word, Microsoft Excel, Microsoft Power Point, Microsoft Access, Page Maker, Coral Draw, Photoshop, Tally etc are some examples.



THANK YOU